

## 5G/6G COMMUNICATION

### Ex.No.4: OFDMA Resource Block Allocation

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**SLOT-A**

**AIM:** To simulate and analyze OFDMA (Orthogonal Frequency Division Multiple Access) resource block allocation and evaluate system performance in terms of resource utilization and user throughput using MATLAB.

#### **Objective**

1. To determine the Number of Resource Blocks (RBs) allocated in an OFDMA system based on available bandwidth and system configuration.
2. To evaluate the Resource Block Utilization (%) by analyzing how efficiently available RBs are assigned to active users.
3. To calculate the User Throughput (Mbps) based on allocated resource blocks and modulation efficiency in the OFDMA system.

#### **MATLAB CODE:**

```
clc; clear; close all;

% System parameters

bandwidth = 10e6; % 10 MHz

rb_size = 180e3; % 180 kHz per RB

% Total Resource Blocks

total_RB = floor(bandwidth / rb_size);

% Users

users = 1:6;

% RB allocation (simple equal + remainder distribution)

base = floor(total_RB / length(users));
```

```

alloc = base * ones(1,length(users));

alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) + 1;

utilization = (alloc / total_RB) * 100;

disp('Total Resource Blocks:')

disp(total_RB)

disp('RB Allocation per User:')

disp(alloc)

figure

% ----- Graph 1: RB Allocation per User -----

subplot(2,1,1)

stem(users, alloc, 'filled', 'LineWidth', 2)

title('OFDMA Resource Block Allocation per User')

xlabel('User Index')

ylabel('Number of Resource Blocks')

grid on

% ----- Graph 2: RB Utilization -----

subplot(2,1,2)

plot(users, utilization, '-o', 'LineWidth', 2)

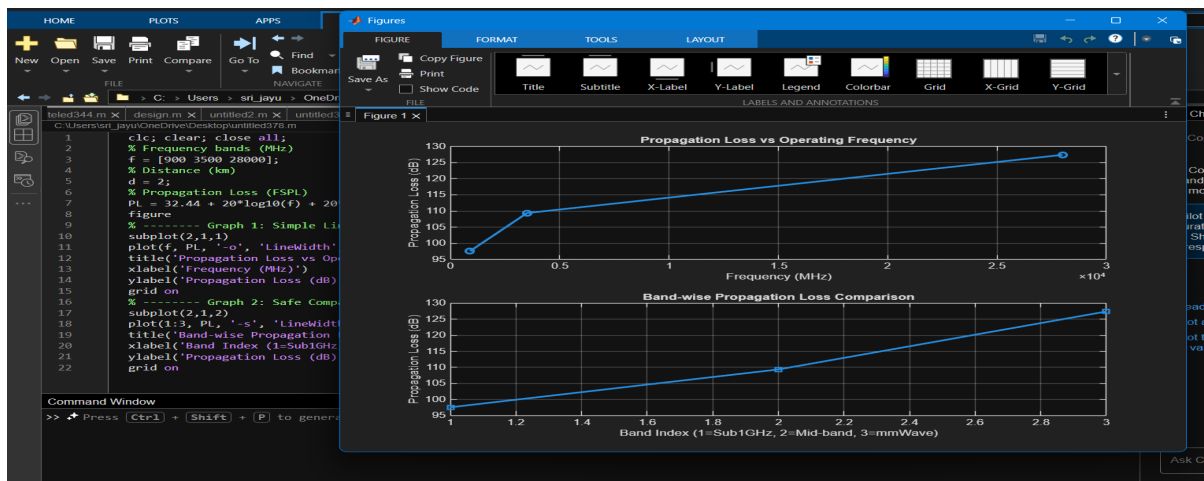
title('OFDMA Resource Block Utilization (%)')

xlabel('User Index')

ylabel('Utilization (%)')

grid on

```



## Objective 2 Matlab Code and Output:

```
clc; clear; close all;

% Total system resource blocks
total_RB = 50;

% Number of active users
users = 1:5;

% RB allocation to users (example scenario)
alloc = [12 10 9 8 11];

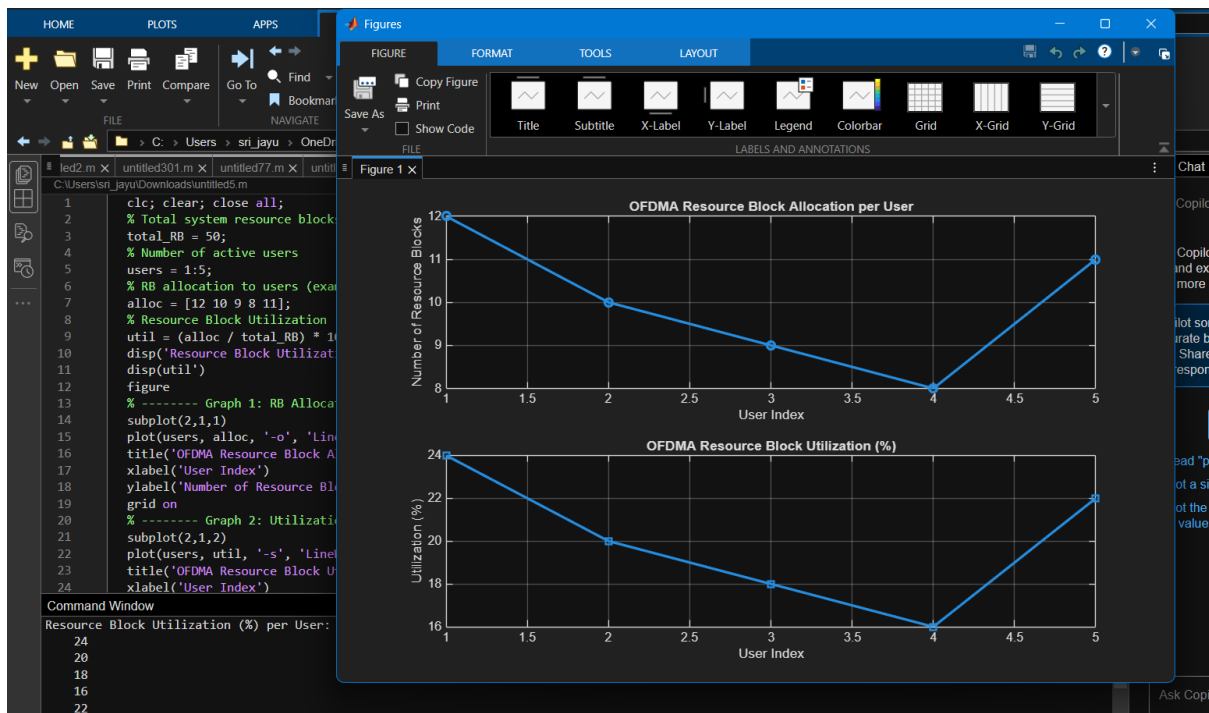
% Resource Block Utilization (%)
util = (alloc / total_RB) * 100;

disp('Resource Block Utilization (%) per User:')
disp(util)

figure

% ----- Graph 1: RB Allocation -----
subplot(2,1,1)
plot(users, alloc, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on

% ----- Graph 2: Utilization (%) -----
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```



### Objective 3 Matlab Code and Output:

```
clc; clear; close all;
```

```
% System parameters
```

```
rb_bw = 180e3; % Bandwidth per RB (180 kHz)
```

```
users = 1:5;
```

```
% Allocated Resource Blocks per user
```

```
alloc_RB = [12 10 9 8 11];
```

```
% Modulation efficiency (bits/symbol)
```

```
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
```

```
% Throughput calculation (Mbps)
```

```
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
```

```
disp('User Throughput (Mbps):')
```

```
disp(throughput')
```

```
figure
```

```
% ----- Graph 1: Throughput per User -----
```

```
subplot(2,1,1)
```

```

plot(users, throughput, '-o', 'LineWidth', 2)

title('OFDMA User Throughput')

xlabel('User Index')

ylabel('Throughput (Mbps)')

grid on

% ----- Graph 2: RB vs Throughput -----

subplot(2,1,2)

plot(alloc_RB, throughput, '-s', 'LineWidth', 2)

title('Resource Blocks vs Throughput Relationship')

xlabel('Allocated Resource Blocks')

ylabel('Throughput (Mbps)')

grid on

```

